

AXIS 0

Operator–SDF–T framework

Enhanced formalization of Tom Klootwijk’s definition

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Central definition

Axis 0 names the domain of **ALL operators** admitted by the framework. Every operator is an operator signed-distance field. In this revision, its signed field value is defined as T , the framework’s temporal coordinate. Every bifurcation carries Klein-bottle field geometry, traversed through bit shifting, one-bit jitter, parity transport, PSI evolution and lowercase φ scaling.

$$\text{Axis } 0 := \mathcal{A}_0, \quad F \in \mathcal{A}_0, \quad T_{F,i}(x) := d_{F,i}(x).$$

Temporal extension. Present-state initialization at $T = 0$; entropy as an explicitly specified grounding observable; “Now” as a constant- T reading interface; and a worked temporal traversal with a seed, transition rule and decoded output.

Revision basis. The original six-page conversation, the previous twelve-page specification, and the newly supplied eight-page continuation. The new continuation’s spacetime interpretation is incorporated through explicit definitions and a separately identified reference profile.

1 Controlling definition and revision map

SOURCE COMMITMENTS AND THE MEANING OF THE NEW EQUATIONS

1.1 Axis 0 means ALL operators

The original controlling statement defines Axis 0 as “all operators themselves defined as signed distance fields” and says that 0 means “ALL operators themselves in the prompts and my formalization.” [S1, p. 5]

$$\text{Axis } 0 := \mathcal{A}_0 = \{\text{operator-fields named or generated in this framework}\}. \quad (1.1)$$

Here 0 is a domain label. It is not a Cartesian axis, a preferred eigenvector, the number of operators, or the statement that every field value is zero. By contrast, $T = 0$ denotes a value of a temporal field at the present/seed boundary.

1.2 Commitments retained and extended

A1 / Operator identity. Every executable constructor, field, evaluator, traversal rule and reader has an operator-field representation in $\mathcal{A}_0$.

A2 / Bifurcation. Every split is realized by a Klein-bottle field object with two addressed outgoing ports and an orientation-parity transition.

A3 / Evolution. Bit-shifted branching, one-bit jitter, PSI, lowercase φ , log-polar lookup, Cantor addressing and kinematic operators remain in the domain.

A4 / Temporal identity. The continuation’s “SDF = T” is adopted as $T_{F,i} = d_{F,i}$, with the same local sign transitions as the SDF.

A5 / Entry and readout. The seed is on $T = 0$; entropy supplies the grounding observable; “Now” selects a specified constant temporal value for decoding.

[S1, pp. 1--2, 5--6; S2, pp. 4--8]

1.3 What version 2.0 adds

Previous specification	Enhanced specification
Operator--SDF domain	Operator--SDF--T identity, including typed temporal evaluation.
Twisted local distance	Temporal parity transport and an orientation-tracked scalar.
Unassigned entry/output	Present-state seed and constant- T decoding contracts.
Unassigned termination	Entropy observable, explicit stopping predicate and finite-hit condition.
Address/parity example	Fully assigned temporal traversal profile and calculated output.

Reading convention. Source statements establish the vocabulary and requested identities. Equations labeled as constructions are choices introduced in this revision. A profile assigns parameters. A derived result follows from those equations. This separates the supplied definition from the implementation choices needed to evaluate it.

2 The closed operator-field domain

RETAINED CONSTRUCTION / FIELD IDENTITY AND EXECUTABLE MEANING

2.1 A field object and its temporal evaluation

Use a field object F with a carrier X_F , metric g_F , zero set Z_F , local signed-distance functions $d_{F,i}$, sign-transition data, and a field-readable operator description. In each distance chart,

$$|d_{F,i}(x)| = \text{dist}_{g_F}(x, Z_F), \quad T_{F,i}(x) := d_{F,i}(x). \quad (2.1)$$

The field F , its operator interpretation, and its time-field interpretation are views of the same admitted object. A scalar evaluation $T_{F,i}(x)$ is a value of that object; it is not the entire operator language.

The equality uses normalized model units: one unit of signed distance is one unit of T . A conversion into externally measured units would require an assigned conversion map; no physical conversion constant is supplied in the source. [S2, pp. 7--8]

2.2 Closure of all admitted operators

Let Σ contain the retained names, let ι realize them as fields, and let Γ contain the admitted finite-arity field constructors. Define

$$A^{(0)} = \iota(\Sigma), \quad A^{(n+1)} = A^{(n)} \cup \Gamma(A^{(n)}), \quad \mathcal{A}_0 = \bigcup_{n \geq 0} A^{(n)}. \quad (2.2)$$

Descriptions may contain reference links. A finite self-referential description need not be expanded into an infinite syntax tree.

Field application has the signature

$$\text{App} : \bigcup_{m \geq 0} (\mathcal{A}_0 \times \mathcal{A}_0^m) \rightarrow \mathcal{A}_0. \quad (2.3)$$

When an application returns, its result is again an operator-field. Numbers, bits, tuples and stores in the equations denote field-represented data. Partiality records the possibility that an evaluation does not return; it does not introduce a separate timeout rule.

2.3 Internality of the new interfaces

Closure includes the temporal interfaces

Seed, Time, Entropy, Halt, Now, Decode, quotation and evaluation are themselves represented in $\mathcal{A}_0$. Describing them with ordinary mathematical notation does not place their implementations outside Axis 0.

A realization assigns the field-readable description format and its interpretation. Distinct operators may share a zero carrier; their readable descriptions must still distinguish their behavior. Neither a shared distance value nor a shared geometric outline alone determines an instruction. [V1, §2; S2, pp. 4--7]

3 Klein-bottle bifurcation fields

RETAINED CONSTRUCTION / EXACT LOCAL DISTANCE AND THE SIGN TWIST

3.1 Carrier construction

On $\mathbb{R}^2 \times \mathbb{R}$, use coordinates (u, v, z) and the isometries

$$A(u, v, z) = (u + 1, v, z), \quad B(u, v, z) = (-u, v + 1, -z). \quad (3.1)$$

Define the quotient carrier and zero section

$$E_3 = (\mathbb{R}^2 \times \mathbb{R}) / \langle A, B \rangle, \quad K = \{[u, v, 0]\} \subset E_3. \quad (3.2)$$

At $z = 0$, one coordinate is periodic and the other identification reverses u ; this is the Klein-bottle square-identification model retained from version 1.0. Use the quotient Euclidean metric.

In a local lift, define

$$d_i([u, v, z]) = z_i, \quad T_i = d_i, \quad |T_i| = \text{dist}_{E_3}([u, v, z], K). \quad (3.3)$$

The distance identity follows directly: a vertical segment reaches K with length $|z|$; every path to K has length at least the required change of $|z|$. The quotient maps preserve $|z|$.

3.2 One-bit orientation transport

Let $h_{ij} \in \mathbb{B} := \{0, 1\}$ indicate whether a chart transition reverses the sign. Then

$$d_j = (-1)^{h_{ij}} d_i, \quad T_j = (-1)^{h_{ij}} T_i, \quad p_j = p_i \oplus h_{ij}. \quad (3.4)$$

The overlap consistency law is $h_{ik} = h_{ij} \oplus h_{jk}$. An orientation-reversing closed route in the zero carrier has total transition parity 1. Traversing it twice gives parity 0.

3.3 How the binary split is attached

At each split address w , use a Klein field $(E_{3,w}, K_w)$ with two ports, labeled 0 and 1. A port is an addressed location or chart, with a map to a child field; it is not a boundary component of the closed Klein surface.

Bifurcation object

$\text{Bif}(F)$ has a Klein zero carrier, exact local distance/time magnitude, a one-bit sign transition, and two addressed outgoing ports. Its field result belongs to $\mathcal{A}_0$.

The tree supplies the address structure; the Klein carrier supplies the local twisted geometry. The source's "temporal Klein Bottle" is realized by assigning $T = d$ to that same field, rather than by substituting a time-loop label for the geometry. [V1, §3; S1, pp. 2, 5--6; S2, p. 8]

4 SDF = T and temporal orientation

NEW CONSTRUCTION / LOCAL TIME, LIFTED TIME AND THE FOUR-COORDINATE MODEL

4.1 Local signed time and orientation-tracked time

A chart reversal changes the sign of T_i . Accordingly, define the orientation-tracked value on a lifted state by

$$\widehat{T}(F, x, p) := (-1)^p T_{F,i}(x). \quad (4.1)$$

Under simultaneous chart and parity transport,

$$(-1)^{p \oplus h} ((-1)^h T_i) = (-1)^p T_i. \quad (4.2)$$

Thus $\widehat{T}$ is independent of the chosen local representative of the lifted state. It is **not** a new field replacing T , and it is not a global signed scalar on the unlifted quotient. It is a consistent reading of T with an orientation sheet attached.

The parity bit controls sign transport. The jitter bit changes a branch choice. Their values need not coincide. Also, changing coordinates at the same state is distinct from moving to a new state whose distance/time value changes.

4.2 Explicit four-coordinate extension

To realize the continuation's four-dimensional description, take the product

$$E_4 = E_3 \times \mathbb{R}_y, \quad Z_4 = K \times \mathbb{R}_y, \quad T_i = z_i. \quad (4.3)$$

The quotient generators act on (u, v, y, z) as

$$A : (u, v, y, z) \mapsto (u + 1, v, y, z), \quad B : (u, v, y, z) \mapsto (-u, v + 1, y, -z). \quad (4.4)$$

Here $|T_i| = \text{dist}_{E_4}(x, Z_4)$. At fixed y , the bifurcation zero carrier remains K . The full zero hypersurface is the family $K \times \mathbb{R}_y$, not a two-dimensional Klein bottle mislabeled as a three-dimensional surface.

For the lifted state space, attach $p \in \mathbb{B}$, let A preserve p , and let B send p to $p \oplus 1$. Quotienting by those extended maps makes $\widehat{T} = (-1)^p z$ well-defined.

4.3 Meaning of a temporal loop

A reversing loop exchanges the local temporal orientation; two reversals restore it. This is the precise parity identity attached to the source's past/future and cause/effect wording. A law for causal influence, physical time travel or paradox resolution is not supplied in the continuation, so those additional claims are not consequences of the parity equation. [S2, pp. 7--8; construction extends V1, §3]

5 Operator vocabulary and syntax

RETAINED TERMINOLOGY / NO UNSTATED SUBSTITUTIONS BETWEEN SYMBOLS

All names below denote field-represented operators or constructors. A name without an evaluation law remains a named parameter; it is not silently identified with a similarly shaped standard symbol. [S1, pp. 1--2; V1, §6]

Source wording	Notation	Meaning retained or required
men T room	men T room	Literal label; expansion, arity and rule are unassigned.
circle; sphere; cone; apex	Circle, . . .	Geometric carriers and field constructors with declared parameters.
legenda; side view of the pyramid	Legenda, SideView	Notation/view operators; geometry, projection and encoding are assigned by a profile.
PSI math +	$\Psi_{\varphi}, +_{\text{PSI}}$	PSI is the transition interface; the source-specific plus law is separate.
plus; encircled T	$+_{\text{src}}, T_{\circ}$	Retained separately from numerical addition, bit XOR and temporal T .
AT; B floppy P	AT, B floppy P	Literal operator names; signatures and evaluations remain parameters.
kindred search tree	Kindred	Tree-related selection or relation; its relation and selection law are required.
double UU set operator	UU	Retained name; no automatic replacement by ordinary union.
0 order computer science	ZeroOrder	Literal source phrase; distinct from Axis 0's ALL-operators definition.
SDF = T; Present; Entropy; Now	Time, Seed, Entropy, Now	New temporal interpretation and explicit interfaces in this revision.

5.1 Arity, precedence and the circled symbol

For a name σ with declared arity m_{σ} , use fully parenthesized expressions

$$e ::= \sigma \mid \sigma(e_1, \dots, e_{m_{\sigma}}). \quad (5.1)$$

An infix rendering requires a declared precedence and associativity table. In the equations here, $\oplus$ means XOR on bits only; $+$ means numerical addition only. Neither notation assigns the meaning of $+_{\text{PSI}}$, $+_{\text{src}}$ or $T_{\circ}$.

Case sensitivity. Lowercase φ is the scaling symbol requested in the source. Uppercase Φ is a different, unused symbol. The new temporal T does not rename or define the original encircled- T operator.

6 Geometric fields and exact scaling

RETAINED CONSTRUCTION / DISTANCE-VALUED TIME UNDER GEOMETRIC TRANSFORMATIONS

6.1 Circle, sphere, cone, apex and side view

For a circle in a Euclidean plane or a sphere in three dimensions, a selected outward-positive field is

$$d_F(x) = \|x - c\| - R, \quad R > 0, \quad T_F(x) = d_F(x). \quad (6.1)$$

The center, radius and chart are operands. The present level $T = 0$ is the corresponding boundary. For a specified solid cone or pyramid, one possible construction is signed distance to its boundary, with the interior-negative convention stated in the profile. Height, opening, cap and apex data are operands, not inferred from the word “cone.”

A side view is a declared projection P of the selected geometry. Its image is view data; a projected image is not automatically an exact SDF. The original circle/sphere/cone carriers and their Klein bifurcation fields remain explicitly connected but distinct geometric roles. [S1, p. 1; V1, §6]

6.2 Exact similarity transport

For scale $s > 0$, orthogonal R , and translation b , define

$$d_{F'}(x) = s d_F(R^T(x - b)/s), \quad Z_{F'} = b + sRZ_F. \quad (6.2)$$

Since R preserves length,

$$\text{dist}(x, Z_{F'}) = \inf_{y \in Z_F} \|x - b - sRy\| \quad (6.3)$$

$$= s \inf_{y \in Z_F} \|R^T(x - b)/s - y\| = |d_{F'}(x)|. \quad (6.4)$$

The output factor s is required for exact distance magnitude. For corresponding points $x' = b + sRx$, the temporal identity gives

$$T_{F'}(x') = sT_F(x). \quad (6.5)$$

On a Klein carrier, apply this rule in compatible local lifts, transport the geometric scale data with the field, and retain its sign cocycle. A separate reversing chart contributes the factor $(-1)^h$.

Scaling a field is not the same operation as advancing a state

Choosing $s = \varphi^\eta$ scales the field and its corresponding temporal values. Moving a point within a fixed field may instead increase T . The reference profile uses a shrinking branch radius and an increasing orientation-tracked time; it does not equate those two operations.

[V1, §7; temporal interpretation added from S2, pp. 7--8]

7 Bit-shifted tree, jitter and parity

RETAINED TRANSITION ALGEBRA / EXPLICIT DISTINCTION BETWEEN EXPLORATION AND LOOKUP

7.1 Address update

Let $\mathcal{T} \subseteq \mathbb{B}^*$ be prefix-closed with root ϵ . At step n , let q_n be the nominal branch, j_n the one-bit jitter, and c_n the executed branch:

$$c_n = q_n \oplus j_n, \quad w_{n+1} = w_n c_n. \quad (7.1)$$

Starting from integer root address $a_0 = 1$, define

$$a_{n+1} = (a_n \ll 1) \mid c_n = 2a_n + c_n. \quad (7.2)$$

The leading 1 retains zero-valued branches in the stored binary word. Right shift retrieves the parent and the least significant bit retrieves the last branch.

A specified law J supplies the jitter stream. One bit specifies the value set $\mathbb{B}$, not a probability distribution. A deterministic profile fixes the stream or generator; a probabilistic profile states its distribution. [S1, p. 2; S2, pp. 2, 4--5]

7.2 Parity update and local time

Let $h_n \in \mathbb{B}$ be the orientation character of the selected port/chart transport. Then

$$p_{n+1} = p_n \oplus h_n. \quad (7.3)$$

For pure re-expression of the same field point, $T_{\text{new}} = (-1)^{h_n} T_{\text{old}}$. For an actual state transition, first apply the specified motion or field update and then evaluate the resulting field. A parity flip alone is not an elapsed-time increment.

On a reversing closed route γ in a Klein carrier,

$$\bigoplus_{e \in \gamma} h(e) = 1. \quad (7.4)$$

This topological condition is separate from merely toggling a bit on an abstract binary tree.

7.3 The ordered-search condition

To call a realization an ordered binary search tree, provide keys κ in a total order, with

$$\kappa(u) < \kappa(w) < \kappa(v) \quad (u \text{ in the left subtree, } v \text{ in the right subtree}). \quad (7.5)$$

Equal keys select the current node. Exact key lookup follows the comparisons. Jittered exploration is a different traversal mode unless the profile supplies a rule that preserves or restores exact lookup. The reference profile in Section 15 is explicitly an exploration/encoding mode. [V1, §4; S2, p. 1]

8 Lowercase phi, log-polar LUT and Cantor addresses

RETAINED FORMULAS / THE GOLDEN-RATIO CHOICE IS EXPLICIT

8.1 Scale and inverse logarithm

Keep lowercase $\varphi > 0$ as a profile parameter; use $\varphi > 1$ for the contraction/expansion convention below. Set

$$r_{n+1} = \varphi^{\eta_n} r_n, \quad \eta_n \in \{-1, +1\}. \quad (8.1)$$

The continuation uses the golden-ratio interpretation. The reference profile adopts it explicitly:

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \varphi^2 = \varphi + 1. \quad (8.2)$$

This specializes version 1.0's free positive scale; it does not retroactively change the original author's unspecified numerical parameter. [S1, pp. 2--3; S2, p. 2; V1, §5]

For $r > 0$, reference $r_* > 0$, base $b > 0$, $b \neq 1$, define

$$\lambda = \log_b(r/r_*), \quad r = r_* b^\lambda, \quad \lambda_{n+1} = \lambda_n + \eta_n \log_b \varphi. \quad (8.3)$$

“Inverse logarithmically” is realized here by the inverse function b^λ , not by $1/\log_b r$. The apex/origin is handled by a separate chart rather than a finite invented value of $\log 0$.

8.2 Lookup indices

On a declared polar chart with $\theta \in [0, 2\pi)$, choose $\Delta\lambda > 0$, offset λ_0 , and angular bin count $M \geq 1$:

$$i_r = \left\lfloor \frac{\lambda - \lambda_0}{\Delta\lambda} \right\rfloor, \quad i_\theta = \left\lfloor \frac{M\theta}{2\pi} \right\rfloor, \quad \text{LUT}(i_r, i_\theta) \in \mathcal{A}_0. \quad (8.4)$$

The actual table values, ordering and their role in Ψ_φ are assigned data. A coordinate formula does not itself supply a lookup table.

8.3 Cantor encoding of a branch word

For an infinite word $c_1 c_2 \dots$, define

$$C(c) = \sum_{k=1}^{\infty} \frac{2c_k}{3^k}. \quad (8.5)$$

For a finite prefix, $C_n = \sum_{k=1}^n 2c_k/3^k$. The tail is between 0 and 3^{-n} , so all completions lie in $[C_n, C_n + 3^{-n}]$. The finite word remains the exact stored address; the series is its geometric address representation.

9 The enhanced PSI transition

NEW INTEGRATED INTERFACE / TEMPORAL EVALUATION IS DERIVED FROM THE FIELD

9.1 State and derived temporal data

Use a state

$$s_n = (n, w_n, a_n, x_n, p_n, F_n, M_n, \xi_n), \quad (9.1)$$

where M_n is the field/tree store and ξ_n is the input, jitter or generator state. Components are field-represented. Define the temporal and entropy observations by

$$T_n = d_{F_n, i_n}(x_n), \quad \widehat{T}_n = (-1)^{p_n} T_n, \quad \mathcal{E}_n = \text{Entropy}_{\Theta}(s_n). \quad (9.2)$$

They are evaluated observations, not independent numbers that can be changed without changing the field state. The profile Θ assigns their evaluation rules.

9.2 One complete transition

Let χ be the nominal decision, J the jitter rule, τ the port transport, h its orientation bit, $\mathcal{U}$ the field update, and $\mathcal{W}$ the store update. Define

$$(j_n, \xi_{n+1}) = J(s_n), \quad q_n = \chi(s_n), \quad c_n = q_n \oplus j_n, \quad (9.3)$$

$$x_{n+1} = \tau_{\Theta}(s_n, c_n), \quad p_{n+1} = p_n \oplus h(s_n, c_n), \quad (9.4)$$

$$F_{n+1} = \mathcal{U}_{\Theta}(s_n, c_n, x_{n+1}, p_{n+1}), \quad (9.5)$$

$$M_{n+1} = \mathcal{W}_{\Theta}(M_n, w_n c_n, F_{n+1}, x_{n+1}, p_{n+1}), \quad (9.6)$$

$$w_{n+1} = w_n c_n, \quad a_{n+1} = 2a_n + c_n. \quad (9.7)$$

The counter becomes $n + 1$. Together these equations define $s_{n+1} = \Psi_{\varphi}(s_n)$ wherever the assigned rules evaluate. A LUT may supply χ , h , field updates or multiple components when the profile specifies that connection.

9.3 Temporal coherence

After the state is updated, reevaluate

$$T_{n+1} = d_{F_{n+1}, i_{n+1}}(x_{n+1}), \quad \widehat{T}_{n+1} = (-1)^{p_{n+1}} T_{n+1}. \quad (9.8)$$

A proposed independent temporal recurrence is admitted only with a field/coordinate transition that realizes it. Section 15 supplies such a realization, rather than appending an unrelated clock to an SDF.

Closed execution

$\mathcal{K}_{\Theta} = (\mathcal{A}_0, \mathcal{S}, \Psi_{\varphi})$ is the operational kernel. Seed, transition, entropy observation, stopping and readout are all described by field-represented rules in this same domain.

[V1, §8; integrated with S2, pp. 6--8]

10 Temporal kinematics and metric data

NEW FORMAL CONNECTION / DERIVATIVES, SCALE AND THE SELECTED METRIC

10.1 Trajectory parameter versus the SDF time value

Let τ be a trajectory parameter and $x(\tau)$ a twice-differentiable local path. It indexes the path; it is not a second identity redefining T . For a twice-differentiable evolving field,

$$v = \frac{dx}{d\tau}, \quad a = \frac{d^2x}{d\tau^2}, \quad \dot{T} = \partial_\tau d_F + \nabla_x d_F \cdot v. \quad (10.1)$$

When $\dot{T} \neq 0$, the path may be reparameterized by its local SDF time:

$$\frac{dx}{dT} = \frac{v}{\dot{T}}, \quad \frac{d^2x}{dT^2} = \frac{a\dot{T} - v\ddot{T}}{(\dot{T})^3}. \quad (10.2)$$

At a sign-changing chart transition, transport T , coordinates and parity together. At a nondifferentiable field point, use the discrete PSI rule rather than assigning a nonexistent classical derivative.

10.2 Lowercase phi and temporal change

For exact similarity transport at corresponding points, $T' = \varphi^n T$. In particular,

$$T_0 = 0, \quad T_{n+1} = \varphi^{\eta_n} T_n \implies T_n = 0 \text{ for all } n. \quad (10.3)$$

Therefore a present seed does not acquire nonzero temporal motion from scaling alone. A spatial displacement, field deformation or an explicitly realized driving term is required. Section 15 uses a normal displacement whose magnitude is the next φ -scaled radius.

For a fixed-branch differentiable spatial update Ξ_φ , eigenvectors are defined by

$$J_n = D_x \Xi_\varphi(x_n), \quad J_n e_k = \mu_k e_k. \quad (10.4)$$

These are eigenvectors of an assigned map. They do not change Axis 0 into an eigenvector-indexed coordinate axis.

10.3 Metric selected for the four-coordinate construction

The explicit E_4 model uses

$$g = du^2 + dv^2 + dy^2 + dz^2 = du^2 + dv^2 + dy^2 + dT_i^2. \quad (10.5)$$

Both quotient generators preserve this metric, including the simultaneous reversal of u and T_i . It is the positive-definite metric used to prove the distance equality. The continuation asks about a metric tensor but does not specify a physical signature, field equation or causal law. Those data would be additional profile choices, not deductions from $T = d$. [S2, pp. 3, 6--8; V1, §7]

11 Entry boundary: the seed is Present

NEW CONSTRUCTION / AN EXACT INITIALIZATION AND INPUT-PRESERVATION CONTRACT

11.1 Seed interface

Let $\mathcal{I}$ be the declared external input format. A field-represented seed operator maps an input to an initial state:

$$\text{Seed}_\Theta : \mathcal{I} \longrightarrow \mathcal{S}, \quad u \longmapsto s_0. \quad (11.1)$$

The present-state conditions are

$$n_0 = 0, \quad w_0 = \epsilon, \quad a_0 = 1, \quad p_0 = 0, \quad x_0 \in Z_{F_0}, \quad T_0 = \widehat{T}_0 = 0. \quad (11.2)$$

The profile additionally initializes its scale, ports, store, jitter input and entropy value. This is the continuation's "Seed Vector" and "Present" expressed as a concrete entry contract. [S2, pp. 6--8]

11.2 Input content cannot be recovered from zero alone

Let $\text{Enc}_0(u)$ be a field-readable description stored in M_0 . For lossless input handling, require

$$\text{Read}_0(\text{Enc}_0(u)) = u. \quad (11.3)$$

Every seed may have temporal value zero while carrying a different readable input description. The scalar $T_0 = 0$ locates the seed on its zero boundary; it does not encode all distinct input strings by itself.

The initial operator field, the input description and the chosen present point together form the seed. A boundary-crossing operator must specify each of them. Assigning a name to the boundary does not supply an omitted byte-to-field encoding.

11.3 Starting temporal motion

One explicit local construction in E_4 is to choose a positive step δ_n , advance along the signed normal direction determined by p_n , and then apply a chart transport:

$$z_{\text{adv}} = z_n + (-1)^{p_n} \delta_n, \quad z_{n+1} = (-1)^{h_n} z_{\text{adv}}, \quad p_{n+1} = p_n \oplus h_n. \quad (11.4)$$

Since $d = z$, this gives

$$\widehat{T}_{n+1} = \widehat{T}_n + \delta_n. \quad (11.5)$$

The first positive δ_0 moves the present seed into nonzero field time without changing the identity $T = d$. Section 15 sets $\delta_n = r_n/\varphi$. This driving law is introduced for that profile; the source does not supply its numerical equation.

Two meanings of zero, kept distinct

Axis 0 names all admitted operators. Present $T = 0$ is the zero level of the active temporal field. Initializing at Present does not remove operators from the domain.

12 Entropy and the grounding/halting rule

NEW CONSTRUCTION / AN OBSERVABLE, A STOPPING PREDICATE AND A FINITE-HIT CONDITION

12.1 Specify the entropy observable

Let $\mathcal{E} : \mathcal{S} \rightarrow \mathbb{R}$ be the profile's entropy or complexity observable. The continuation calls entropy the halting metric; its exact operational form here is a supplied observable together with a predicate. [S2, pp. 6, 8]

For a forward-entropy profile, impose

$$\mathcal{E}_{n+1} - \mathcal{E}_n \geq 0. \quad (12.1)$$

This is an assigned transition property, not a consequence of the phrase “one-bit jitter.” The source supplies neither a thermodynamic entropy functional nor a probability law for the jitter. The reference profile therefore uses an explicitly defined combinatorial entropy.

12.2 Exact completion semantics

A threshold profile assigns $\mathcal{E}_*$, declares its decoder on terminal states, and sets

$$H_{\Theta}(s) = \mathbf{1}\{\mathcal{E}(s) \geq \mathcal{E}_*\}, \quad N = \inf\{n \geq 0 : H_{\Theta}(s_n) = 1\}. \quad (12.2)$$

A run returns at the first such state. If no such n exists, that run has no terminal result. No unrelated depth limit, elapsed-time cutoff or extra stopping condition is implicit in this definition. A different grounding rule must be stated in its profile.

12.3 When finite termination actually follows

Assume each preterminal state has a defined next step. If $\mathcal{E}_0 < \mathcal{E}_*$ and every preterminal step obeys

$$\mathcal{E}_{n+1} - \mathcal{E}_n \geq \varepsilon > 0, \quad (12.3)$$

then after at most

$$N \leq \left\lceil \frac{\mathcal{E}_* - \mathcal{E}_0}{\varepsilon} \right\rceil \quad (12.4)$$

steps the threshold is reached. The proof is the telescoping bound $\mathcal{E}_n \geq \mathcal{E}_0 + n\varepsilon$.

Monotonicity alone is insufficient: $\mathcal{E}_n = n/(n+1)$ increases strictly but never reaches $\mathcal{E}_* = 1$ in a finite number of steps. Likewise, a velocity approaching zero or a constant parity bit needs an explicit stopping test before it defines a terminal state. These are consequences of the stated tests, not added runtime mechanisms.

12.4 Jitter, direction and entropy are separate assignments

A deterministic jitter stream can change routes without creating a probability distribution. A stochastic stream requires such a distribution. To identify jitter with an entropy increment or a temporal arrow, the profile must supply the coupling equation. The reference profile fixes all three streams, advances $\widehat{T}$, and counts prefix capacity explicitly.

13 The slicing interface: Now

NEW CONSTRUCTION / CONSTANT TEMPORAL VALUE, ORIENTATION AND READABLE OUTPUT

13.1 A well-defined constant-time slice

On the orientation-tracked carrier $\widehat{E}_4$, define

$$\text{Now}(c) = \{\widehat{x} \in \widehat{E}_4 : \widehat{T}(\widehat{x}) = c\}. \quad (13.1)$$

The lifted value is used because local T changes sign at a reversing chart transition. On the unlifted carrier, $|T| = |c|$ is an invariant alternative, but it does not distinguish the two temporal orientations.

In the construction of Section 4, select representatives with $p = 0$. The remaining identifications are $u \mapsto u + 1$ and $v \mapsto v + 2$, with y, z unchanged. Thus

$$\widehat{E}_4 \cong \mathbb{T}^2 \times \mathbb{R}_y \times \mathbb{R}_{\widehat{T}}, \quad \text{Now}(c) \cong \mathbb{T}^2 \times \mathbb{R}_y. \quad (13.2)$$

Here $\mathbb{T}^2 = (\mathbb{R}/\mathbb{Z}) \times (\mathbb{R}/2\mathbb{Z})$. Each Now slice is three-dimensional; $du \wedge dv \wedge dy$ supplies a consistent orientation in this explicit lifted model. The projected zero slice is $K \times \mathbb{R}_y$, so its topology is not silently equated with that of the lift.

Locally $\widehat{T} = c$ is a coordinate hyperplane. Globally it is the glued level set above. “Slicing Hyperplane” is retained as the interface name, not as a claim that every nonlinear field level set is a flat Euclidean plane. [S2, pp. 7--8]

13.2 Reading marked fields rather than only distances

A finite-state realization selects a finite collection of field-readable marked records from M_n . Define

$$P_c(s) = \{(a, x, p, \text{payload}) \in M_s : \widehat{T}(x, p) = c\}, \quad (13.3)$$

with the selection and ordering specified by the profile. Then

$$\text{ReadNow}_{\Theta}(s, c) = \text{Codec}_{\Theta}(\text{Order}_{\Theta}(P_c(s))). \quad (13.4)$$

The payload may encode text, bytes, geometry or another field. Its codec determines the output format. An image or sampled asset additionally needs specified sample locations and rendering rules; a continuous level set by itself is not a serialized file.

13.3 Terminal readout

For a completed run, select $c = \widehat{T}_N$ and return $\text{ReadNow}_{\Theta}(s_N, c)$. The reference profile makes $\widehat{T}_n$ strictly increasing, so the terminal slice contains a unique terminal record. Equality at its stored time is exact in the mathematical profile; approximate implementations must declare their comparison rule.

14 Self-reference, replication and universal execution

RETAINED OPERATIONAL CONTRACTS / EXTENDED TO TEMPORAL ENTRY AND READOUT

14.1 Self-referential evaluation

Let $Q(F) \in \mathcal{A}_0$ be a field-readable description with $\text{Read}(Q(F)) = F$. A field evaluator satisfies

$$\text{Eval}(Q(F), G) \simeq \text{App}(F; G). \quad (14.1)$$

The equivalence means the same terminating result or nontermination on both sides. For a field F_Ψ representing PSI,

$$\text{Eval}(Q(F_\Psi), \text{Enc}(s)) \simeq \text{Enc}(\Psi_\varphi(s)). \quad (14.2)$$

This is an operational self-reference equation. A geometric loop does not replace the quotation, reader or evaluator needed to realize it.

14.2 Self-replication

For a field-readable kernel description D_Θ , a replication rule must produce fresh addresses $v_0 \neq v_1$ satisfying

$$\text{Rep}(D_\Theta, M) \Downarrow (M', v_0, v_1), \quad \text{Read}(M'[v_0]) \equiv \text{Read}(D_\Theta) \equiv \text{Read}(M'[v_1]). \quad (14.3)$$

The relation $\equiv$ means the same specified behavior, including seed, temporal transition, grounding and decoder rules. Different positions or φ -scales are allowed when the reader preserves that behavior. Producing two addressed records is not, by itself, copying an executable kernel.

14.3 Universal simulation

Fix an effective machine encoding and write $M_p(u)$ for program p on input u . A universal profile supplies computable input/output encodings E, D and a field program U such that

$$D(\text{Run}_\Theta(U, E(p, u))) \simeq M_p(u) \quad \text{for every } p, u. \quad (14.4)$$

The run includes its seed, stopping observation and Now readout. A probabilistic profile states the exact probability mode of the equivalence.

The supplied conversations describe self-replication and universality, but do not give the encodings, evaluator, replication construction or simulation proof required by these equations. They remain explicit realization properties, rather than conclusions inferred from topology or entropy alone. [S1, pp. 3--6; S2, pp. 3--5; V1, §§8--9]

Classification. An engine executes a selected set of operations; its kernel is the state-transition mechanism. Self-reference, self-replication and universality are the additional contracts above. A finite entropy-terminated traversal does not establish universal simulation.

15 Reference profile R: temporal bit-stream traversal

NEW, FULLY ASSIGNED EXAMPLE / A CONCRETE REALIZATION OF THE TEMPORAL INTERFACES

15.1 Input, seed and fields

The input is three bit words Q, J, H of equal finite length L . They supply nominal branches, jitter and port-orientation bits respectively. This profile explores and encodes a path; it is not exact ordered-key lookup.

Use $\varphi = (1 + \sqrt{5})/2$, the carrier E_4 , local field $d = z$, and input-readable marked records. Seed

$$(u_0, v_0, y_0, z_0) = (0, 0, 1, 0), \quad r_0 = y_0 = 1, \quad p_0 = 0, \quad a_0 = 1, \quad w_0 = \epsilon. \quad (15.1)$$

The input words are stored in a quoted tuple in M_0 ; the reader returns that tuple. This marked-field description is the profile's input encoding. At every visited address, the active field has the same temporal law $d = z$ and its own readable record.

15.2 One temporal transition

For $0 \leq n < L$, use $q_n = Q[n]$, $j_n = J[n]$, $h_n = H[n]$, and set

$$c_n = q_n \oplus j_n, \quad p_{n+1} = p_n \oplus h_n, \quad r_{n+1} = r_n / \varphi, \quad (15.2)$$

$$u_{n+1} = (-1)^{h_n} u_n, \quad v_{n+1} = v_n + h_n, \quad y_{n+1} = r_{n+1}, \quad (15.3)$$

$$z_{n+1} = (-1)^{h_n} (z_n + (-1)^{p_n} r_{n+1}), \quad (15.4)$$

$$a_{n+1} = 2a_n + c_n, \quad w_{n+1} = w_n c_n. \quad (15.5)$$

Both ports at depth n are assigned the orientation character $H[n]$. The u, v, z change realizes the actual Klein generator when $h_n = 1$, following the stated normal advance; it is not an unattached parity toggle.

With $T_n = z_n$, these equations give

$$\widehat{T}_{n+1} = \widehat{T}_n + r_{n+1}, \quad r_n = \varphi^{-n}, \quad \widehat{T}_n = \frac{1 - \varphi^{-n}}{\varphi - 1}. \quad (15.6)$$

Store one marked record (a_n, x_n, p_n, w_n) per visited event.

15.3 Entropy, lookup and output

Define combinatorial prefix-capacity entropy $\mathcal{E}_n = \log_2 |\mathbb{B}^n| = n$, and stop when $\mathcal{E}_n = L$. It counts available length- n branch words, not thermodynamic entropy or measured randomness. For lookup, use the local tangential Cartesian pair (y, u) ; on this trace, $r = \sqrt{y^2 + u^2} = y$ and $\theta = 0$. Set $b = \varphi$, $r_* = 1$, $\lambda_n = -n$, $\Delta\lambda = 1$, $M = 1$, $\lambda_0 = 0$, and define the finite LUT row $(-n, 0)$ to return the quoted triple $(Q[n], J[n], H[n])$ for $n < L$.

Read Now($\widehat{T}_L$), select its record, and serialize w_L as an ASCII bit string. The threshold L is the input's exact completion condition, not an unrelated iteration cap.

16 Reference transition algorithm and local matrix

EXECUTABLE PSEUDOCODE / EXACT ARITHMETIC SEMANTICS

The following algorithm is the scalar/record realization of profile R. The marked records represent fields with $T = d = z$; the code does not implement the unassigned source operators in Section 5.

```

INPUT: Q, J, H are bit words of equal length L
phi := (1 + sqrt(5)) / 2
n := 0; a := 1; p := 0; word := ""
u := 0; v := 0; y := 1; z := 0
store := [record(a, (u,v,y,z), p, word)]

while n < L:
    (q, j, h) := LUT(-n, 0)      # quoted Q[n], J[n], H[n]
    c := q XOR j
    y_next := y / phi
    z_next := (-1)^h * (z + (-1)^p * y_next)
    u := (-1)^h * u
    v := v + h
    p := p XOR h
    y := y_next; z := z_next
    a := 2*a + c
    word := word concatenated with bit(c)
    n := n + 1
    store.append(record(a, (u,v,y,z), p, word))

now := (-1)^p * z
terminal := the record with lifted_time(record) = now
RETURN ASCII(terminal.word)

```

For $L = 0$, the seed is already terminal and the returned word is empty. For nonempty input, each lifted-time increment is positive, so the terminal selector is unique.

16.1 Exact local transition matrix

For fixed p, h , put $\epsilon_h = (-1)^h$. The affine coordinate update has Jacobian, in the order (u, v, y, z) ,

$$J_{p,h} = \begin{pmatrix} \epsilon_h & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \varphi^{-1} & 0 \\ 0 & 0 & \epsilon_h(-1)^p \varphi^{-1} & \epsilon_h \end{pmatrix}. \quad (16.1)$$

Its eigenvalues are the diagonal entries: $\epsilon_h, 1, \varphi^{-1}, \epsilon_h$. For example, the coordinate vectors in the u, v , and z directions are eigenvectors. This supplies a definite kinematic matrix; it does not imply contraction in every direction.

Arithmetic. Since $\varphi^{-1} = \varphi - 1$, finite-step radii, coordinates and times can be represented exactly in $\mathbb{Q}(\sqrt{5})$. Displayed decimal values in the next section are rounded presentations, not the definition of equality used for slicing.

17 Worked temporal trace and decoded output

CALCULATED EXAMPLE / THE PREVIOUS BRANCH TRACE EXTENDED BY SDF = T

Choose the same decision, jitter and parity-transition inputs used in the previous specification:

$$Q = 0110, \quad J = 1010, \quad H = 1011, \quad L = 4. \quad (17.1)$$

The executed branch word is $Q \oplus J = 1100$. The initial entropy, local time and lifted time are all zero. [V1, §10; temporal quantities calculated from profile R]

Step	q	j	c	h	a	p	$\mathcal{E}$
0	--	--	--	--	1	0	0
1	0	1	1	1	3	1	1
2	1	0	1	0	7	1	2
3	1	1	0	1	14	0	3
4	0	0	0	1	28	1	4

Step	Radius r_n	Local time $T_n = z_n$	Lifted time $\widehat{T}_n$
0	1.000000	0.000000	0.000000
1	0.618034	-0.618034	0.618034
2	0.381966	-1.000000	1.000000
3	0.236068	1.236068	1.236068
4	0.145898	-1.381966	1.381966

17.1 Exact terminal quantities

$$w_4 = 1100, \quad a_4 = 28 = (11100)_2, \quad p_4 = 1, \quad (17.2)$$

$$r_4 = \varphi^{-4} = 5 - 3\varphi, \quad \lambda_4 = -4, \quad \mathcal{E}_4 = 4, \quad (17.3)$$

$$\widehat{T}_4 = 3 - \varphi, \quad T_4 = \varphi - 3, \quad (u_4, v_4) = (0, 3). \quad (17.4)$$

The local time is negative because the current parity is 1. The lifted time is positive and has increased at every step. Their relation remains $T_4 = (-1)^{p_4} \widehat{T}_4$.

The Cantor-prefix coordinate and completion interval remain

$$C_4 = \frac{2}{3} + \frac{2}{9} = \frac{8}{9}, \quad [C_4, C_4 + 3^{-4}] = \left[\frac{72}{81}, \frac{73}{81} \right]. \quad (17.5)$$

Readout at Now

$\mathcal{E}_4 = L$, so the run terminates at step 4. On $\text{Now}(3 - \varphi)$, the unique terminal record has address 28 and payload 1100. The decoder returns **1100**, encoded as ASCII bytes 31 31 30 30 in hexadecimal.

18 Derived properties and conformance checks

ALGEBRAIC CONSEQUENCES / WHAT THE ASSIGNED PROFILE ESTABLISHES

18.1 Address and parity invariants

After n branch decisions,

$$a_n = 2^n + \sum_{k=1}^n c_{k-1} 2^{n-k}, \quad p_n = p_0 \oplus \bigoplus_{k=0}^{n-1} h_k. \quad (18.1)$$

Both follow by induction from the step equations. The address is recoverable, and two sign reversals cancel. All generated records remain in the field closure because the record constructor is an admitted field constructor.

18.2 Temporal coherence and completion in profile R

Substituting the update for z and p gives

$$(-1)^{p_{n+1}} z_{n+1} = (-1)^{p_n} z_n + r_{n+1}. \quad (18.2)$$

Therefore $T_n = d(x_n)$ and $\widehat{T}_n = (-1)^{p_n} T_n$ remain true, and $\widehat{T}_{n+1} > \widehat{T}_n$. The selected entropy increases by exactly 1, so the exact input-completion threshold is reached at $N = L$.

For unit step parameter, the temporal increments and their change are

$$\Delta \widehat{T}_n = \varphi^{-(n+1)}, \quad \Delta \widehat{T}_{n+1} - \Delta \widehat{T}_n = (\varphi^{-1} - 1) \varphi^{-(n+1)}. \quad (18.3)$$

This gives a forward discrete temporal velocity whose increments decrease in magnitude. It is an equation of the chosen profile, not an inferred physical law.

18.3 Bounded time is distinct from finite termination

For an unending extension of the same radius/advance recurrence,

$$\widehat{T}_n \longrightarrow \frac{1}{\varphi - 1} = \varphi, \quad \mathcal{E}_n = n \longrightarrow \infty. \quad (18.4)$$

Thus infinitely many mathematical steps can fit below a finite limiting field-time value. Profile R terminates because its finite input length is the specified threshold, not because the time series is bounded or because parity changes. This makes the exact role of the source's entropy proposal explicit.

Conformance check	Required result
Seed and input	$T_0 = 0$; reading the stored input returns (Q, J, H) .
Distance and orientation	$ d = \text{dist}(x, Z_4)$; chart sign and parity transformations agree.
Step coherence	Address, parity, coordinates and derived T satisfy the same step.
Entropy completion	$\mathcal{E}_n = n$; first terminal state occurs at $n = L$, including $L = 0$.
Now and decoder	The terminal lifted-time record is unique and returns the branch word.
Four-step reference	Address 28, parity 1, entropy 4, $\widehat{T} = 3 - \varphi$, output 1100.

19 Realization parameter register

ASSIGNED VALUES, RETAINED INTERFACES AND SOURCE GAPS

The enhanced definition is a family of realizations indexed by Θ . Profile R fixes one finite traversal. The remaining source vocabulary is preserved without fabricating an instruction set for it.

Component	General realization	Reference profile R
Field language	Encoding, reader, constructors and evaluator in $\mathcal{A}_0$.	Marked $d = z$ fields and quoted bit/record tuples; no general evaluator assigned.
SDF/time identity	$T_{F,i} = d_{F,i}$; carrier, metric and local sign data specified.	E_4 , Euclidean quotient metric, $T = z$, lifted value $(-1)^p z$.
Seed/Present	Input codec, present point, initial store and jitter state.	Three finite bit words; $(0, 0, 1, 0)$, root address 1, parity 0.
Klein ports	Port maps, node data and orientation character.	Generator B after normal advance when $H[n] = 1$; two ports per split.
Tree mode	Key map for ordered search, or declared exploration rule.	Exploration; $c_n = Q[n] \oplus J[n]$.
Scale and LUT	$\varphi, b, r_*, \eta, \Delta\lambda, M$, bins and actual entries.	Golden ratio; contraction; $b = \varphi$; row $(-n, 0)$ reads the input triple.
Temporal dynamics	Field/coordinate law realizing any proposed time increment.	$\delta_n = r_n/\varphi$, with the signed normal update in Section 15.
Entropy and halt	Functional, increment law, terminal predicate and decoder domain.	$\mathcal{E}_n = n$, threshold L , exact finite-input completion.
Now/output	Oriented slice or magnitude slice, record selection, ordering and codec.	Lifted slice $\widehat{T}_L$; unique event record; ASCII branch word.
Source-specific names	Arity and laws for T_o , +PSI, AT, UU, B floppy P, Kindred and men T room.	Names retained; they are not invoked by the example.
Replication / universality	Kernel description, copy rule, universal program, encodings and simulation.	Contracts retained; the finite traversal is not a realization of these extra properties.
Physical interpretation	Units, causal dynamics, entropy functional and any physical metric choice.	Normalized mathematical time-field; no external physical identification assigned.

No hidden interpretation. Supplying a value for one entry does not determine the others. In particular, $T = d$ alone does not determine the seed codec, entropy functional, temporal forcing, decoder or universal evaluator. Each has an explicit place in this register.

19.1 Calculation record

The scalar reference recurrence was checked in exact $\mathbb{Q}(\sqrt{5})$ arithmetic for all 4,681 triples of equal-length bit words with lengths 0 through 4, and for 100 additional triples of length 32. Address recovery, accumulated parity, SDF/time coherence, the closed temporal formula and unique terminal-event selection passed in all **4,781 cases**. These checks concern the assigned reference recurrence and its decoder.

20 Source lineage and revision record

SOURCE-GROUNDED CONSOLIDATION / DEFINITIONS, CONSTRUCTIONS AND CALCULATED RESULTS

20.1 Document basis

[S1] **Original six-page conversation.** Uploaded PDF beginning “men T room a circle a sphere a cone an apex legenda a side view of the pyramid lower case phi log encoded polar LUT binary search tree klein bottle signed distance fields.” The author-supplied prompts appear alongside accompanying explanations. Page 1 supplies the vocabulary; page 2 adds jitter, lowercase phi, B floppy P and bifurcation; page 3 asks about engine/kernel/framework classification; page 5 states the controlling ALL-operators meaning of Axis 0.

[V1] **Previous specification.** *Axis 0: Operator--SDF framework*, twelve pages, version 1.0 (the previously delivered formal specification). Its operator closure, exact local distance construction, branching equations, symbol register and operational contracts form the retained mathematical base.

[S2] **New eight-page continuation.** Uploaded hyphenated-filename PDF, from men-T-room-through -klein-bottl.pdf, containing exported Google Search response text. Pages 1--5 reprise the domain vocabulary and closure; page 6 introduces seed, grounding and decoding; page 7 contains the entry-to-output flow and the SDF-to-T diagram; page 8 develops Present, entropy, Now and the temporal Klein-bottle interpretation.

[R] **Source prompt and current request.** Personal identifiers are reproduced exactly from the source prompt. The current request asks for an enhanced PDF incorporating the new attachment. Explanatory claims in the attachments are distinguished from author-issued definitions and from derived mathematical results.

20.2 Where the new source material is incorporated

Source location	Material incorporated	Sections
S2, pp. 4--5	Axis 0 domain, operator/field identity and closed evolution.	1--2, 9
S2, pp. 6--7	Seed, grounding, decoder and the complete input/output sequence.	11--13
S2, pp. 7--8	SDF = T, Present and constant-T “Now.”	2--4, 10--13
S2, p. 8	Temporal Klein bottle and entropy-based completion.	3--4, 12
New constructions	Explicit four-coordinate/lifted model, driven profile, matrix, temporal trace and conformance tests.	4, 10, 13, 15--18

20.3 Identity record

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Version 2.0 consolidates the previous formalization with the new temporal material. It preserves the requested names and identities while assigning a concrete, source-distinguished seed-to-output profile whose equations can be evaluated directly.